

Operator 1: Pure-Algebraic Mathematical Specification for the Local Graph Expansion Operator $\mathcal{G}_{n \rightarrow n+1}$

1. Unified Mathematical Notation Index

- $\mathbb{N}^+$: Set of positive integers $\{1, 2, 3, \dots\}$ (zero excluded).
 - $\mathcal{I}_n$: Index-baseline set $\{1, 2, 3, \dots, n\}$. All global slicing and matrix-element access strictly adopt **1-based indexing**.
 - $\mathbf{1}_n$: n -dimensional all-ones column vector.
 - $\mathcal{M}_n$: n -th-order discrete-valued square-matrix space (set of purely binary symmetric real matrices).
 - $\mathcal{V}_k$: Set of formal symbolic variables introduced at the k -th evolution frontier.
 - $\mathcal{R}_\infty$: Inductive-limit ring constructed over canonical-embedding maps for multivariate-polynomial rings.
 - Φ : Global evaluation-homomorphism mapping defined over $\mathcal{R}_\infty$.
 - Φ_{full} : Full-connection fixed-evaluation homomorphism (all-ones special case).
 - $\mathcal{G}_{n \rightarrow n+1}$: Local Graph Expansion Operator (declarative dimension-extension operator).
 - $\rho(A)$: Spectral radius of matrix A (maximum absolute eigenvalue).
 - $\varinjlim$: Inductive-limit (direct-limit) operator in commutative-algebra category theory.
 - $\partial_{n \rightarrow m}$: Canonical-embedding operator extending polynomial rings from order n to order m .
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2. Algebraic Construction of Inductive-Limit Ring and Global Homomorphism

To handle formal variables introduced by successive expansions within a single algebraic object, we adopt the inductive-limit mechanism from commutative-algebra category theory and define scenario-isolation constraints: formal-symbol operations reside in polynomial-ring spaces; numerical evaluations take place over real numbers. Homomorphism preservation (additivity and multiplicativity) separates these two domains.

2.1 Formal-Variable Isolation, Equivalence Judgement and Limit-Ring Construction

Upon system expansion from step k to $k + 1$, the newly-introduced formal-variable set is strictly denoted:

$$\mathcal{V}_{k+1} = \{x_{(k+1,1)}, x_{(k+1,2)}, \dots, x_{(k+1,k)}, y_{k+1}\}.$$

For $k \neq m$, we have $\mathcal{V}_k \cap \mathcal{V}_m = \emptyset$.

Let $\mathcal{R}_n = \mathbb{R}[\bigcup_{i=2}^n \mathcal{V}_i]$ be the finite multivariate-polynomial ring.

Introduce the family of canonical-embedding maps

$$\partial_{n \rightarrow m} : \mathcal{R}_n \hookrightarrow \mathcal{R}_m \quad (\forall n \leq m \in \mathbb{N}^+).$$

This family satisfies transitivity-compatibility over the directed system:

$$\partial_{m \rightarrow k} \circ \partial_{n \rightarrow m} = \partial_{n \rightarrow k} \quad (\forall n \leq m \leq k).$$

Define the full historical formal-symbol ring as the category-theoretic **inductive limit** of this directed-inclusion system:

$$\mathcal{R}_\infty = \varinjlim \mathcal{R}_n = \left(\bigoplus_{n=1}^{\infty} \mathcal{R}_n \right) / \sim$$

Equivalence-relation rule ($\sim$) for formal polynomials inside this limit ring:

Given two symbolic polynomials $f \in \mathcal{R}_a, g \in \mathcal{R}_b$ belonging to finite-order rings, f and g are equivalent ($f \sim g$) **if and only if** there exists a sufficiently large common evolution-order $N \geq \max(a, b)$ such that their images under canonical embeddings inside the higher-order sub-ring $\mathcal{R}_N$ are identical:

$$\partial_{a \rightarrow N}(f) \equiv \partial_{b \rightarrow N}(g) \quad \text{within } \mathcal{R}_N.$$

2.2 Global Evaluation Homomorphism and Matrix-Space Definitions

Based on limit ring $\mathcal{R}_\infty$, under unconditional real-symmetry constraints ($A^T = A, B^T = B$), we define algebraic spaces and morphisms:

1. **Discrete-valued square-matrix space** ($\mathcal{M}_n \subseteq \{-1, 1\}^{n \times n}$): Matrix space of purely binary real scalars. Its top-left sub-matrix unconditionally inherits constant numerical values from the prior step and contains no indeterminates.
2. **Formal symbolic square-matrix space** ($\mathcal{M}_{n+1}[\mathbf{x}_{n+1}, y_{n+1}] \subseteq \left(\mathbb{R}[\mathcal{V}_{n+1}] \right)^{(n+1) \times (n+1)}$): Space of formal-polynomial parameter matrices.
3. **Global evaluation homomorphism** ($\Phi : \mathcal{R}_\infty \rightarrow \mathbb{R}$): Algebraic morphism preserving addition and multiplication. Hard range constraints apply for all historical or current-step variables:

$$\Phi(x_{(k,m)}) \in \{-1, 1\} \quad (\forall k \in \mathbb{N}^+, \forall m \in \mathcal{I}_{k-1}), \quad \Phi(y_k) \in \{-1, 1\} \quad (\forall k \in \mathbb{N}^+).$$

3. Graph-Theory Terminology and General Topological-Frustration Criterion Theorems

3.1 Formal Walks and Path-Interference Definitions

- **Formal symbolic graph walk**: Formal matrix power multiplication inside polynomial-matrix space. Matrix element $(M^k)_{ij}$ denotes the formal expanded polynomial for weighted paths connecting node i to node j .
- **Path interference**: When multiple formal paths are summed linearly, this yields destructive sign-opposite cancellation or constructive same-sign superposition of polynomial terms.
- **Topological frustration**: After mapping via global homomorphism Φ , due to interleaved network-loop structures, under any assignment satisfying discrete-range constraints the local net-bias absolute value cannot be reduced to zero.

3.2 General Algebraic Criteria for Topological Frustration

Derived to adaptively judge generalized multi-loop systems admitting self-loops and multiple edges.

Theorem 1: Single-Circuit Topological-Frustration Criterion

Theorem summary: A single circuit inside a generalized graph is topologically frustrated **if and only if** the discrete algebraic product over all edge weights along that closed loop equals -1 .

Let realized matrix $M_n \in \mathcal{M}_n$ correspond to generalized graph G . Suppose G contains a closed loop of length L with vertex sequence $v_1 \rightarrow v_2 \rightarrow \dots \rightarrow v_L \rightarrow v_1$. The necessary-and-sufficient condition for this circuit to be frustrated reads:

$$\prod_{m=1}^{L-1} M_n(v_m, v_{m+1}) \cdot M_n(v_L, v_1) = -1.$$

Corollary 1: Global Algebraic-Frustration-Basis Theorem for Complex Superposed Systems

Theorem summary: A multi-loop system with self-loops is globally frustration-free **if and only if** every generator in its generalized cycle-space basis yields product $+1$.

Define the generalized cycle-space basis for a discrete-algebra multi-loop graph system, spanned jointly by fundamental independent circuits plus every vertex self-loop. The system is globally free of topological frustration iff both conditions hold:

1. **Fundamental-circuit condition:** Product of edge weights for every independent fundamental circuit in the basis equals $+1$.
2. **Self-loop condition:** Every vertex self-loop satisfies $M_n(i, i) = 1 \quad (\forall i \in \mathcal{I}_n)$.

If any generator evaluates to -1 , algebraic independence of basis generators forbids cancelling this local negative bias; global frustration necessarily arises.

3.3 Numerical-Matrix Demonstration for Complex Multi-Loop Frustration

Consider this third-order real-valued matrix:

$$M_3 = \begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & -1 \\ 1 & -1 & -1 \end{pmatrix} \in \mathcal{M}_3.$$

Test against generalized cycle-space basis:

1. Evaluate product for fundamental circuit $1 \rightarrow 2 \rightarrow 3 \rightarrow 1$:
 $M_3(1, 2) \cdot M_3(2, 3) \cdot M_3(3, 1) = 1 \cdot (-1) \cdot 1 = -1$. Violates condition $1 \rightarrow$ circuit frustrated.
2. Inspect diagonal self-loops: $M_3(2, 2) = -1$. Violates condition $2 \rightarrow$ self-loop conflict.

By the algebraic-frustration-basis theorem, basis generators contain -1 ; the whole system falls into deep frustration.

Evaluate the two-step-walk path-interference entry $(M_3^2)_{23}$ via row-column inner-product expansion:

$$\begin{aligned} (M_3^2)_{23} &= M_3(2, 1)M_3(1, 3) + M_3(2, 2)M_3(2, 3) + M_3(2, 3)M_3(3, 3) = \\ &= 1 \cdot 1 + (-1)(-1) + (-1)(-1) = 3. \end{aligned}$$

The result yields deterministic real scalar 3. Numerically demonstrates residual local net-bias arising from multi-loop superposition and self-loop sign entanglement which cannot be cancelled to zero.

4 Rigorous Mapping Specification and Property Proof for Local Graph Expansion Operator

4.1 Iteration Origin and Operator Mapping

Absolute iteration origin is constant matrix $M_1 = (1) \in \mathcal{M}_1$.

Full name: Local Graph Expansion Operator; notation: $\mathcal{G}_{n \rightarrow n+1}$.

It acts strictly upon one discrete realized square matrix with mapping:

$$\mathcal{G}_{n \rightarrow n+1} : \mathcal{M}_n \longrightarrow \mathcal{M}_{n+1}[\mathbf{x}_{n+1}, y_{n+1}].$$

4.2 Matrix-Expansion Structural Equation and Explicit-Vector Definition

For any input realized matrix $M_n \in \mathcal{M}_n$, the expansion operator outputs a unique formal polynomial block-matrix:

$$\mathcal{G}_{n \rightarrow n+1}(M_n) = \begin{pmatrix} M_n & \mathbf{x}_{n+1} \\ \mathbf{x}_{n+1}^T & y_{n+1} \end{pmatrix}.$$

Frontier-coupling formal column vector:

$$\mathbf{x}_{n+1} = \begin{bmatrix} x_{(n+1,1)} \\ x_{(n+1,2)} \\ \vdots \\ x_{(n+1,n)} \end{bmatrix} \in (\mathbb{R}[\mathcal{V}_{n+1}])^n.$$

The formal symbolic matrix enforces read-only subspace inheritance:

$$M_{n+1}[1:n, 1:n] \equiv M_n.$$

The $n + 1$ algebraic indeterminates inside frontier set $\mathcal{V}_{n+1}$ are mutually algebraically independent.

4.3 Walk-through Example: Primordial Expansion $M_1 \rightarrow M_2$

1. **Input initial state:** Starting constant $M_1 = (1) \in \mathcal{M}_1$.
2. **Operator invocation:** Call $\mathcal{G}_{1 \rightarrow 2}(M_1)$. Declare second-order formal-variable set $\mathcal{V}_2 = \{x_{(2,1)}, y_2\}$.
3. **Output formal structure:** Substitute into structural equation ($n = 1$, vector degenerates to scalar component). Resulting second-order formal matrix:

$$M_2(x_{(2,1)}, y_2) = \begin{pmatrix} 1 & x_{(2,1)} \\ x_{(2,1)} & y_2 \end{pmatrix} \in \mathcal{M}_2[\mathbf{x}_2, y_2].$$

This example illustrates how operator preserves fixed historical constants in top-left block while appending parameterized rows-columns at frontier.

4.4 Operator-Mapping-Properties Proof (Injective but Non-Surjective)

Theorem 2: Injectivity and Non-Surjectivity of Expansion Operator

Theorem summary: Strict injectivity guarantees lossless inheritance of historical heterogeneity; non-surjectivity means its image only covers formal matrices whose top-left sub-block contains purely numerical constant entries.

Proof:

1. Injectivity:

Take two distinct input matrices $M_n^{(1)} \neq M_n^{(2)} \in \mathcal{M}_n$. There exists some index pair $\exists i, j \in \mathcal{I}_n$ such that $M_n^{(1)}(i, j) \neq M_n^{(2)}(i, j)$. By read-only-subspace inheritance property:

$$\mathcal{G}_{n \rightarrow n+1}(M_n^{(1)})[i, j] = M_n^{(1)}(i, j) \neq M_n^{(2)}(i, j) = \mathcal{G}_{n \rightarrow n+1}(M_n^{(2)})[i, j].$$

Hence $\mathcal{G}_{n \rightarrow n+1}(M_n^{(1)}) \neq \mathcal{G}_{n \rightarrow n+1}(M_n^{(2)})$. Injectivity is proven.

2. Non-surjectivity:

Co-domain space $\mathcal{M}_{n+1}[\mathbf{x}_{n+1}, \mathbf{y}_{n+1}]$ in principle permits formal indeterminates inside its top-left $n \times n$ sub-block. Whereas any element from image $\text{Im}(\mathcal{G}_{n \rightarrow n+1})$ must have purely numerical constant top-left sub-block.

Consider matrix B in co-domain such that $B[1, 1] = x_{(n+1,1)}$. This violates constant-sub-block inheritance constraint and cannot belong to operator image. Non-surjectivity proven. ■

5 Divergence Quantification and General-Invariant Proof

5.1 Quantitative Divergence under Full-Connection Fixed Evaluation

If intermediate formal states skip downstream operator solving and are directly forced into full-connection fixed-assignment special case ($\Phi_{\text{full}}: \Phi_{\text{full}}(\mathbf{x}_{n+1}) = \mathbf{1}_n, \Phi_{\text{full}}(\mathbf{y}_{n+1}) = 1$).

Define total-sum series for evaluated real-matrix entries:

$$S_n = \sum_{i=1}^n \sum_{j=1}^n \Phi_{\text{full}}(M_n(i, j)),$$

spectral-radius invariant: $\rho(\Phi_{\text{full}}(M_n))$.

Recurrence relation for global total-sum via block-matrix expansion:

$$S_{n+1} = S_n + 2 \sum_{k=1}^n \Phi_{\text{full}}(x_{(n+1,k)}) + \Phi_{\text{full}}(y_{n+1}).$$

Substitute full-connection assignment:

$$S_{n+1} = S_n + 2n + 1.$$

Starting from initial-condition $S_1 = 1$:

$$S_n = 1 + \sum_{k=1}^{n-1} (2k + 1) = n^2 \implies \lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} n^2 = \infty.$$

Under this assignment M_n reduces to all-ones real matrix $J_n \in \{1\}^{n \times n}$. Its spectral-radius:

$$\rho(\Phi_{\text{full}}(M_n)) = n \implies \lim_{n \rightarrow \infty} \rho(\Phi_{\text{full}}(M_n)) = \infty.$$

Divergence conclusion: Forcing fixed full-connection assignments at expansion frontier yields global scalar-sum diverging in $O(n^2)$ quadratic order and spectral-radius diverging linearly $O(n)$. This justifies algebraically keeping formal-symbol matrices unevaluated, to be resolved only by downstream operators.

5.2 General Diagonal-Invariant Theorem (Proof for arbitrary n)

Theorem 3: General Diagonal-Invariant Theorem

Theorem summary: Within formal-matrix-square setting, after applying binary-assignment homomorphism Φ , diagonal path-interference polynomial for newly injected node evaluates deterministically to constant $n + 1$.

Important scope limitation: This invariant holds **strictly for two-step graph walks only** (M^2). For higher-order powers M^k , $k \geq 3$, cross-terms mixing frontier indeterminates and historical constants appear; diagonal entries no longer reduce to assignment-independent constants.

Proof:

Formally multiply block-structured formal matrix:

$$M_{n+1}^2 = \begin{pmatrix} M_n & \mathbf{x}_{n+1} \\ \mathbf{x}_{n+1}^T & y_{n+1} \end{pmatrix} \begin{pmatrix} M_n & \mathbf{x}_{n+1} \\ \mathbf{x}_{n+1}^T & y_{n+1} \end{pmatrix}.$$

Extract last diagonal entry of product matrix:

$$(M_{n+1}^2)_{n+1, n+1} = \left(\sum_{m=1}^n M_{n+1}(n+1, m)^2 \right) + M_{n+1}(n+1, n+1)^2.$$

Substitute formal-symbol components:

$$(M_{n+1}^2)_{n+1, n+1} = \left(\sum_{m=1}^n x_{(n+1, m)}^2 \right) + y_{n+1}^2.$$

Apply global evaluation homomorphism Φ :

$$\Phi((M_{n+1}^2)_{n+1, n+1}) = \left(\sum_{m=1}^n \Phi(x_{(n+1, m)})^2 \right) + \Phi(y_{n+1})^2.$$

Hard binary-range constraints:

$$\Phi(x_{(n+1, m)}) \in \{-1, 1\}, \quad \Phi(y_{n+1}) \in \{-1, 1\}.$$

Any element squared in this binary-set equals real constant 1.

$$\Phi(x_{(n+1, m)})^2 \equiv 1, \quad \Phi(y_{n+1})^2 \equiv 1.$$

Substitute constants back to summation:

$$\Phi((M_{n+1}^2)_{n+1, n+1}) = \left(\sum_{m=1}^n 1 \right) + 1 = \underbrace{1+1+\dots+1}_{n \text{ terms}} + 1 = n + 1.$$

This reduction holds for arbitrary positive integer $n \in \mathbb{N}^+$, independent of specific assignments for indeterminates. Theorem 3 proven. ■

6 Constraint Compatibility and Iterative-Complexity-Bound Analysis

6.1 Constraint-Compatibility Analysis

Operator 1 outputs formal-symbol matrix containing $n + 1$ algebraically-independent indeterminates; resulting Boolean solution-space size is 2^{n+1} .

Because expansion $\mathcal{G}_{n \rightarrow n+1}$ only linearly adds degrees-of-freedom, and top-left block M_n is numerically fixed in prior iterations, constraint systems built from downstream operators cannot retroactively alter historical fixed solutions. This guarantees **permanent constraint compatibility**: at least one valid binary real-solution always exists for local-interference polynomial systems.

6.2 Iterative-Complexity-Bound Analysis

Separate algebraic-logic complexity versus physical runtime-implementation overhead:

1. **Formal-algebra construction complexity**: Operator 1 only performs formal declaration for unknown-variable block structure; pure formal-logic overhead is strictly constant complexity $O(1)$.
2. **Physical-storage-complexity remark**: The above $O(1)$ applies **only to symbolic-declaration logical layer**. If runtime performs real physical-memory allocation, constructs variable-hash lookup tables or iterates variable-sets, physical-memory overhead scales linearly as $O(n)$.
3. **Downstream formal-matrix-multiplication complexity**: If downstream code computes M_{n+1}^2 formal product, naive multivariate-polynomial expansion upper-bound reads $O((n+1)^3)$. But leveraging that top-left block M_n is purely numerical constant, block-aware symbolic-multiplication reduces practical polynomial-multiplication overhead to $O(n^2)$. This guarantees engineering feasibility for iterating to large system-sizes.

Appendix Directory

For stable large-scale distributed-computation deployment, this algebraic specification defines these auxiliary appendices (accessible via main system catalogue):

- **Appendix A: Topological-Routing Protocols for Causal Data Streams**: Detailed specification for lossless serialization-transmission protocols of parameter-symbol matrices across multi-level data buses.
- **Appendix B: Bidirectional-Conversion Specification between Formal-Polynomial Matrices and Evaluated Real-Valued Matrices**: Defines hash-evaluation mapping tables for homomorphism Φ plus hardware-aware memory-alignment permutation mechanisms.
- **Appendix C: Graph Degradation and Exception-Handling for Empty-Solution Cases**: Roll-back and

topological self-healing algorithm invoked under extreme conditions (e.g. external tampering over read-only blocks) leading to no valid real-solutions for characteristic equations.

- **Appendix D: Downstream-Operator Polynomial-Interface Output-Format Standard:** Layout standard for multivariate-polynomial coefficient matrices when passing off-diagonal path-interference elements toward numerical solvers.